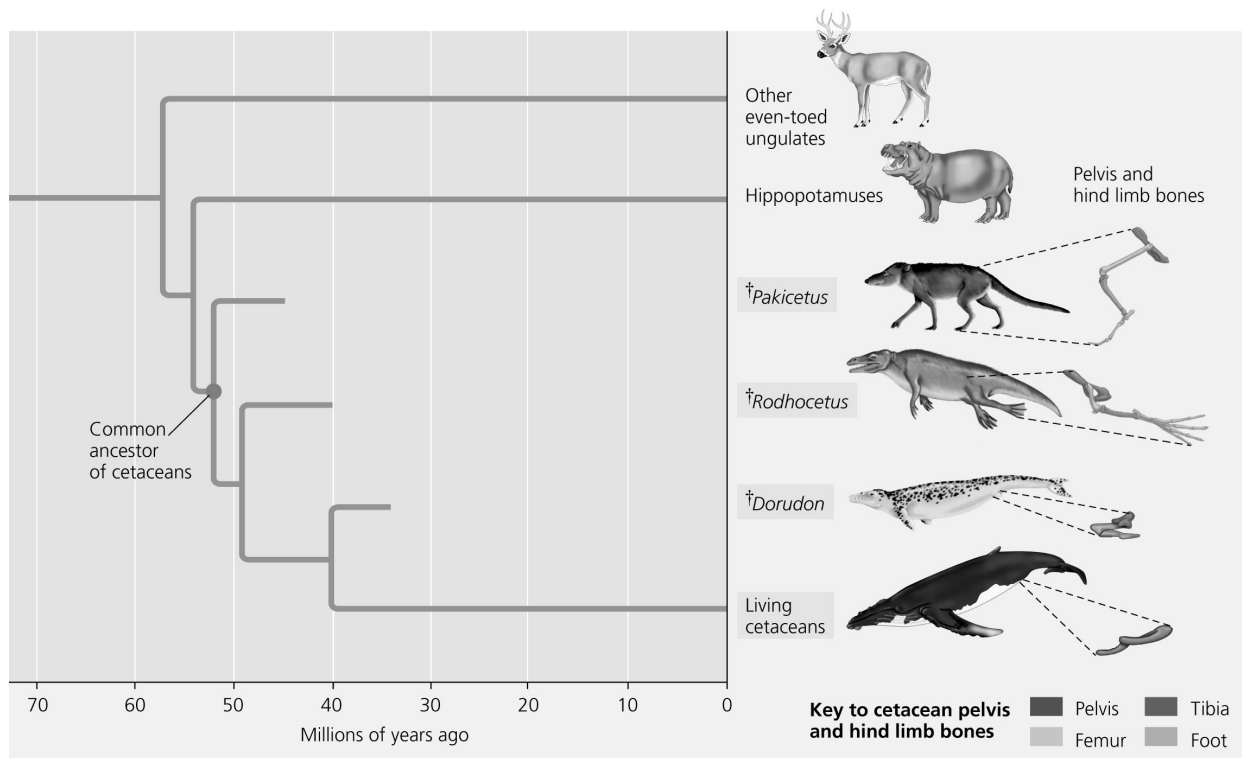


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) Imagine that you are an explorer who has discovered an isolated land mass in the middle of the Pacific Ocean. You find a predator that resembles a North American wolf except that it lays eggs. You name it the "mid-ocean wolf." The physical similarity between the mid-ocean wolf and the North American wolf is most likely due to which of the following factors?
 - A) inheritance of the characteristics from a recent common ancestor
 - B) natural selection that favors the evolution of homologous structures
 - C) natural selection that favors similar traits for organisms with similar ecological roles
 - D) similarities in prey species
- 2) Which of the following observations helped Darwin shape his concept of descent with modification?
 - A) Fewer species live on islands than on the nearest continents.
 - B) Birds live on islands located farther from the mainland than the birds' maximum nonstop flight distance.
 - C) Species diversity declines farther from the equator.
 - D) South American temperate plants are more similar to the tropical plants of South America than to the temperate plants of Europe.
- 3) DNA sequences in many human genes are very similar to the sequences of corresponding genes in chimpanzees. The most likely explanation for this result is that _____.
 - A) chimpanzees evolved from humans
 - B) convergent evolution led to the DNA similarities
 - C) humans and chimpanzees share a relatively recent common ancestor
 - D) humans evolved from chimpanzees
- 4) The upper forelimbs of humans and bats have fairly similar skeletal structures, whereas the corresponding bones in whales have very different shapes and proportions. However, genetic data suggest that all three kinds of organisms diverged from a common ancestor at about the same time. Which of the following is the most likely explanation for these data?
 - A) Natural selection in an aquatic environment resulted in significant changes to whale forelimb anatomy.
 - B) Whales are not properly classified as mammals.
 - C) Genes mutate faster in whales than in humans or bats.
 - D) Forelimb evolution was adaptive in people and bats, but not in whales.
- 5) Which of the following is *not* an observation or inference on which natural selection is based?
 - A) Only a fraction of an individual's offspring may survive.
 - B) There is heritable variation among individuals.
 - C) Poorly adapted individuals never produce offspring.
 - D) Species produce more offspring than the environment can support.

- 6) Within six months of effectively using methicillin to treat *S. aureus* infections in a community, all new *S. aureus* infections were caused by MRSA. How can this best be explained?
- A) Some drug-resistant bacteria were present at the start of treatment, and natural selection increased their frequency.
 - B) In response to the drug, *S. aureus* began making drug-resistant versions of the protein targeted by the drug.
 - C) *S. aureus* evolved to resist vaccines.
 - D) A patient must have become infected with MRSA from another community.
- 7) Use the diagram to answer the next question



- Fossil evidence indicates that modern whales and other cetaceans (dolphins and porpoises) are closely related to even-toed ungulates (hippopotamuses, pigs, deer, and cows). Which of the following predictions would you make if you wanted to test this idea using the amino acid sequence of hemoglobin?
- A) Hemoglobin sequences of whales will be more similar to hippopotamuses than to a fossil common ancestor of all even-toed ungulates (assume we can discover the amino acid sequence from a fossil).
 - B) Hemoglobin sequences of whales will be more similar to dogs than to hippopotamuses.
 - C) Hemoglobin sequences of whales will be more similar to hippopotamuses than to horses.
 - D) Hemoglobin sequences of whales will be more similar to gorillas than to hippopotamuses.
- 8) Which of the following statements is accurate if we view evolution both as a pattern and a process?
- A) To study evolution as a pattern we draw on information from many other scientific fields, while to study evolution as a process we study factors that cause changes in populations.
 - B) To study evolution as a pattern we study data from current day populations, while to study evolution as a process we rely on data from the past.
 - C) To study evolution as a pattern we rely on data from the past, while to study evolution as a process we study data from current day populations.
 - D) To study evolution as a pattern we study factors that cause changes in populations, while to study evolution as a process we draw on information from many other scientific fields.

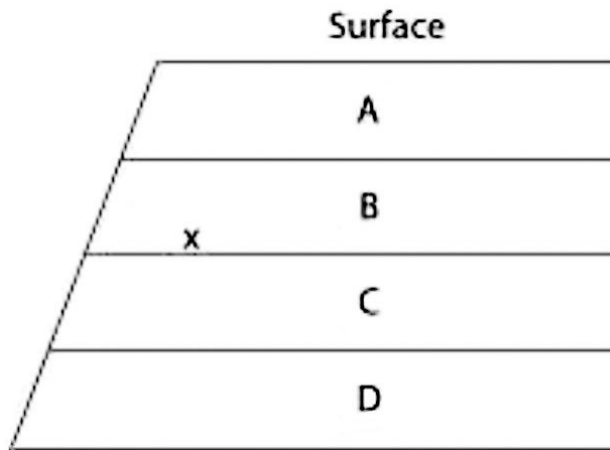
9) Darwin used the phrase "descent with modification" to explain which of the following ideas?

- A) the unity of life
- B) the diversity of life
- C) both the unity and diversity of life
- D) descent of all organisms from a single, ancient ancestor

10) Which of the following statements is most consistent with data obtained from the fossil record?

- A) Geologic changes occur quickly on Earth.
- B) Older strata carry fossils that differ greatly from living organisms.
- C) Unused body parts decrease in size.
- D) Organisms have an innate drive to increase in complexity.

11) The figure shows an outcrop of sedimentary rock whose strata are labeled A-D.



If x indicates the earliest appearance of fossils of two closely related species, neither of which is extinct, then their remains may be found in which of these strata?

- A) strata A and B only
- B) stratum A only
- C) strata A, B, and C only
- D) all four strata

12) James Hutton and Charles Lyell provided which of the following ideas that paved the way for Darwin's thinking?

- A) the surface features of Earth were formed by catastrophic events and those events are still occurring
- B) the surface features of Earth were formed slowly and gradually and the geological processes are still at work
- C) the surface features of Earth were formed by catastrophic events and the fossil record showed that species have evolved
- D) the surface features of Earth were formed gradually and the fossil record showed that species have evolved

13) Which of the following factors contribute to limiting the capacity of organisms to overreproduce?

- A) inability of individuals to tolerate the physical conditions, lack of food, and mortality from predators
- B) lack of food, mortality from predators, and production of many offspring per pair of parents
- C) inability of individuals to tolerate the physical conditions, production of many offspring per pair of parents, and abundance of food
- D) abundance of food, mortality from predators, and inability of individuals to tolerate the physical conditions

- 14) Of the following anatomical structures, which is most homologous to the bones in the wings of a bird?
- A) bony rays in the tail fin of a flying fish
 - B) bones in the flippers of a whale
 - C) chitinous struts in the wings of a butterfly
 - D) bones in the hind limbs of a kangaroo
- 15) Which of the following statements about vestigial structures is accurate?
- A) A vestigial structure in species A can be analogous to a functional structure in Species B.
 - B) A vestigial structure in species A did not have a function in past ancestors.
 - C) A vestigial structure in species A can be homologous to a functional structure in species B.
 - D) A vestigial structure in species A is likely an example of convergent evolution with a structure in species B.
- 16) Which of the following statements best describes Linnaeus' method of classifying organisms?
- A) Linnaeus grouped organisms into increasingly more inclusive categories.
 - B) Linnaeus grouped organisms according to the linear hierarchy of the *scala naturae*.
 - C) Linnaeus grouped organisms into increasingly more complex categories.
 - D) Linnaeus grouped organisms according to their geographical location.
- 17) Continental drift (plate tectonics) has moved continents around the globe. Which of the following predictions would be most likely?
- A) No species will occur on all continents.
 - B) The most recently separated continents will have the most closely related species.
 - C) There is not enough information to make a prediction.
 - D) Most species will occur only on one continent.
- 18) A species of mouse that is preyed on by owls (which can see color) at dawn and dusk varies in coat color between dark and light brown. Under which of the following conditions will NO evolution occur if climate change results in the soil color becoming darker?
- A) Dark-colored mice are more likely to be eaten by owls.
 - B) There is no variability in the genes that control coat color.
 - C) Light-colored mice produce more offspring than dark-colored mice.
 - D) There is no way to make a prediction.
- 19) Which of the following statements about evolutionary trees is accurate?
- A) Evolutionary trees are considered evolutionary theories.
 - B) Evolutionary trees are developed when they are confirmed by fossil evidence.
 - C) Evolutionary trees are considered evolutionary hypotheses.
 - D) Evolutionary trees cannot be tested by gathering more evidence.
- 20) Over long periods of time, many cave-dwelling organisms have lost their eyes. Tapeworms have lost their digestive systems. Whales have lost their hind limbs. How can natural selection account for these losses?
- A) Under particular circumstances that persisted for long periods, natural selection reduced each of these structures because they presented greater costs than benefits.
 - B) The ancestors of these organisms experienced harmful mutations that forced them to lose these structures.
 - C) Natural selection accounts for these losses by the principle of use and disuse.
 - D) Natural selection cannot account for losses, but can only account for new structures and functions.

Answer Key

Testname: UNTITLED1

- 1) C
ID: bio12-- 22.1-54
- 2) D
ID: bio12-- 22.2-2
- 3) C
ID: bio12-- 22.2-5
- 4) A
ID: bio12-- 22.2-4
- 5) C
ID: bio12-- 22.2-1
- 6) A
ID: bio12-- 22.2-3
- 7) C
ID: bio12-- 22.1-50
- 8) A
ID: bio12-- 22.1-4
- 9) C
ID: bio12-- 22.1-25
- 10) B
ID: bio12-- 22.1-3
- 11) A
ID: bio12-- 22.1-7
- 12) B
ID: bio12-- 22.1-9
- 13) A
ID: bio12-- 22.1-24
- 14) B
ID: bio12-- 22.1-39
- 15) C
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- 16) A
ID: bio12-- 22.1-2
- 17) B
ID: bio12-- 22.1-55
- 18) B
ID: bio12-- 22.1-15
- 19) C
ID: bio12-- 22.1-48
- 20) A
ID: bio12-- 22.1-41